

APEX: Automated Pipeline for Explorative Drug Discovery

A target-agnostic QSAR screening and molecular docking platform

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Computational Drug Discovery

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Outline

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The Drug Discovery Challenge

Scale of the Problem

- 12–15 years from target to approval
- \$2.6 billion average per approved drug
- ~1 in 10,000 screened compounds reaches trials

Computational Acceleration

Early-stage virtual screening reduces the wet-lab search space from millions to tens of candidates before any synthesis is attempted.

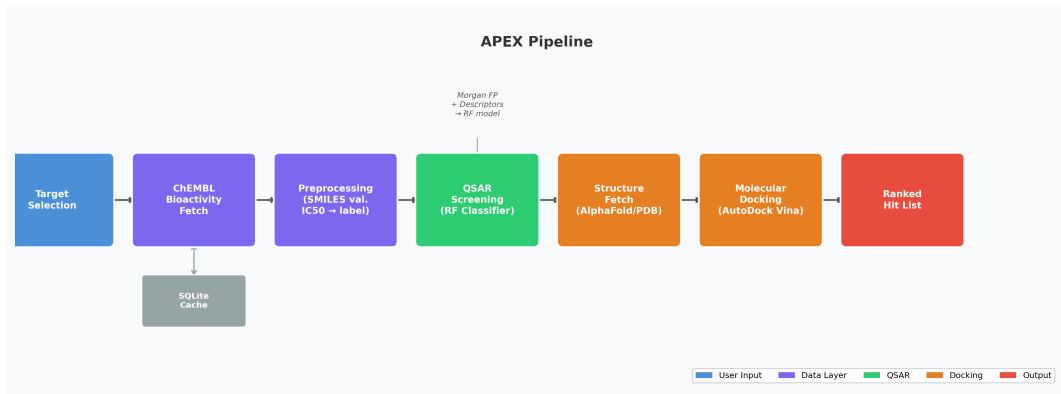
Bottleneck

Manual, target-specific pipelines require expert setup for every new biological target — not scalable.

Our Approach

APEX automates the full pipeline: data ingestion → QSAR → docking for *any* ChEMBL target, on demand.

APEX End-to-End Workflow



Each stage is independently cached and testable; the web platform exposes every stage through a REST API.

ChEMBL Bioactivity Layer

ChEMBL Database

- >19 million bioactivity records
- >2 million unique compounds
- Queried via REST API per target on demand

SQLite Cache

All API responses are cached locally with a 7-day TTL. Subsequent runs for the same target return in milliseconds.

Data Filtering Pipeline

- 1 Fetch IC50 measurements (human assays, nM units)
- 2 Validate SMILES with RDKit MolFromSmiles
- 3 Label: $IC_{50} \leq 1000 \text{ nM} \Rightarrow$ **active**
- 4 Stratified 80/20 train/test split

Rate Limiting

ChEMBL enforces rate limits. Retry logic with exponential backoff (3 attempts, 5 s base) prevents silent data loss.

QSAR Model Architecture

Molecular Featurisation

Morgan Fingerprints (radius = 2, 1024 bits)

Encode circular substructure neighbourhoods up to 2 bonds.

RDKit Descriptors (4 features concatenated):

- MolWt, LogP
- # H-bond donors (HBD)
- # H-bond acceptors (HBA)

Combined feature vector: $\mathbf{x} \in \mathbb{R}^{1028}$

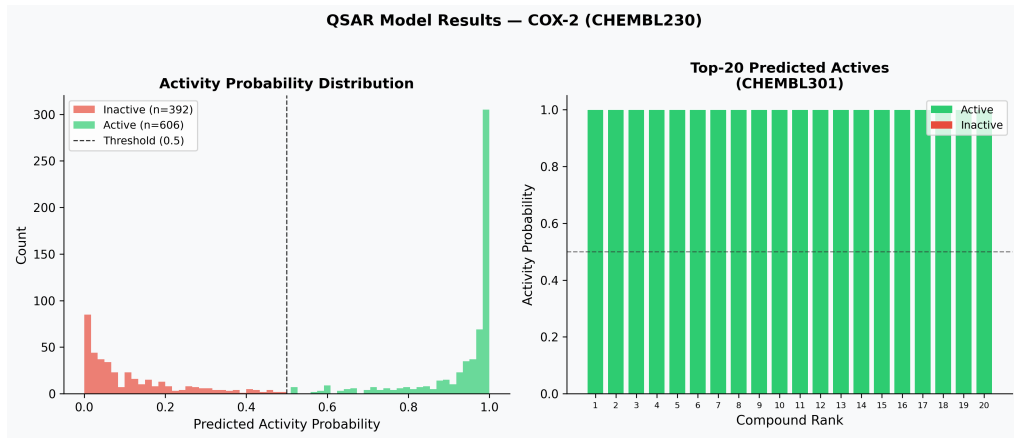
Random Forest Classifier

- 500 estimators
- 5-fold stratified cross-validation
- Reports ROC-AUC and F1 on held-out test set

Output

Per-compound predicted activity probability $p \in [0, 1]$. Top-20 compounds (p ranked descending) forwarded to docking.

QSAR Screening Results



Left: predicted activity probability distributions by class. *Right:* top-20 ranked compounds with active/inactive labels. CDK2 (CHEMBL301) — CV AUC = 0.943, Test F1 = 0.901.

AutoDock Vina Integration

Ligand Preparation

- 1 SMILES $\rightarrow$ 3D conformer (RDKit ETKDGv3)
- 2 Energy minimisation (MMFF94 force field)
- 3 Conversion to PDBQT (Open Babel)

Receptor Preparation

- 1 Structure fetch: AlphaFold $\rightarrow$ RCSB PDB
- 2 Conversion to PDBQT
- 3 Binding box: BioPython C_{α} centroid + 10 Å padding

Vina Scoring Function

Combines:

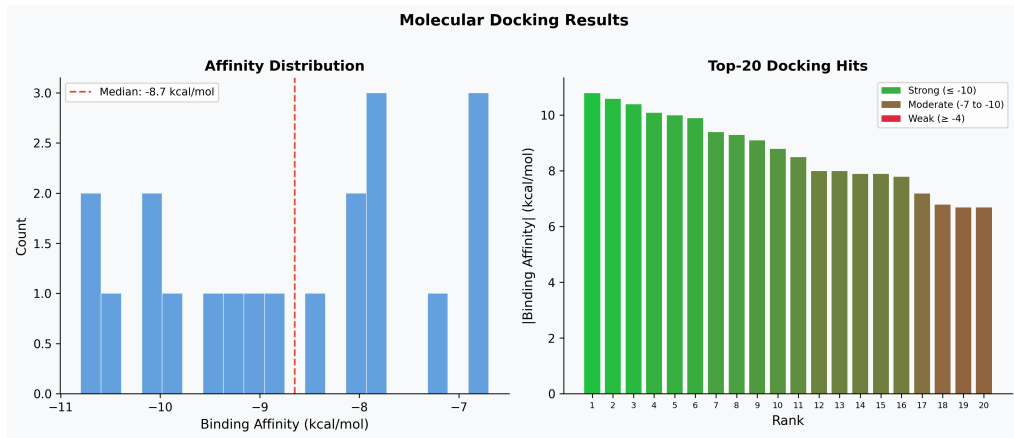
- Steric (Gaussian) terms
- Hydrogen-bond energy
- Hydrophobic contacts
- Torsional entropy penalty

Output: ΔG_{bind} in kcal/mol

Exhaustiveness

Set to 8 (default). Increase for larger binding pockets at the cost of runtime.

Docking Results



Left: full affinity distribution across screened candidates. *Right:* top-20 ranked hits coloured by binding strength (green ≤ -10 kcal/mol, yellow moderate, red ≥ -4 kcal/mol).

Case Study — Cyclin-Dependent Kinase 2 (CDK2)

Target Profile

ChEMBL ID CHEMBL301

UniProt P24941

Function Cell cycle regulator (G1/S transition)

Disease link Cancer (overexpression in many tumours)

Records 998 IC50 measurements retrieved

QSAR Outcome

- CV ROC-AUC: **0.943** $\pm$ 0.012
- Test F1: **0.901**
- Training time: < 30 s

Docking — Top Hit

Best binding affinity: **−10.8 kcal/mol**

Affinity range of top-20: −10.8 to −6.5 kcal/mol

Compounds with $\Delta G < -9$ kcal/mol:
6/20

Total Pipeline Runtime

Stage	Time
ChEMBL fetch	~110 s
QSAR training	<30 s
Docking (20 lig.)	~5 min

Backend — FastAPI

- GET /targets/search?q= — ChEMBL target lookup
- POST /qsar/{target_id} — train + predict
- GET /structures/{uniprot_id} — AlphaFold/PDB fetch
- POST /docking/ — Vina run, ranked results

All routes are independently callable; each service is independently cacheable and testable.

Frontend — React

- Sidebar target search
- Real-time pipeline status log
- QSAR results table with activity probabilities
- Docking hit ranking with affinity heat colours
- Compound compare overlay

No Build Step

Single `index.html` with CDN React — open directly in a browser or serve with any static file server.

Summary & Future Work

What APEX Delivers

- Fully automated, target-agnostic screening pipeline
- ChEMBL bioactivity ingestion with local caching
- QSAR model (RF, AUC > 0.94 on CDK2)
- AutoDock Vina docking for top-N candidates
- REST API + interactive web frontend

Planned Extensions

- Larger virtual compound libraries (Enamine REAL)
- Ensemble QSAR models (XGBoost, GNN)
- ADMET property filtering pre-docking
- 3D binding pose visualisation in-browser
- Multi-target comparative screening

APEX: an open, extensible platform that turns a target name into a ranked hit list — fully automatically.

Questions?

Source code available on request.

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